

R4 — ML Research Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: week 4 analysis commands are ready and understood.

Steps: 1. Read research/tinystories/analyze.py fully. 2. Document what the analyzer refuses and why: missing arms, wrong seed count, mixed cohorts. 3. Write the week-4 commands into a note. One command must be:

```
uv run python -m research.tinystories.analyze --tag cleandeploy --expect-arms baseline,ple,fatembed --expect-seeds 2
```

4. Run the command with an empty runs folder. Record the error message.

Deliverable: the analyzer notes with the error record.

Verify: the analyzer refuses an empty cohort with a clear error.

Learning this week

The MLP forward pass. Write the numpy forward pass for the XOR MLP. Print the hidden activations.

Resources

- analyze.py source: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/analyze.py>
- research README: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/README.md>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Done when

- [] Analyzer notes exist with the error record
- [] The analyzer refuses an empty cohort
- [] You can say what the project does in one sentence